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### APPARATUS AND METHOD FOR PERFORMING COMMUNICATION BY BASE STATION AND DISTRIBUTED ANTENNA SYSTEM

#### Abstract

A method performed by a first distributed antenna system (DAS) node unit in a communication system including a DAS includes: measuring information related to a delay of at least one second DAS node unit connected to the first DAS node unit; generating delay information for the at least one second DAS node unit based on the information related to the delay; and transmitting the delay information to an open-radio access network (O-RAN) node unit of a base station to which the DAS is connected.

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## **Background/Summary**

### **CROSS-REFERENCE TO RELATED APPLICATIONS**

[0001] This application claims the benefit of Korean Patent Application No. 10-2024-0000235, filed on Jan. 2, 2024, in the Korean Intellectual Property Office, the disclosure of which is incorporated herein in its entirety by reference.

### **BACKGROUND**

#### **1. Field**

[0002] The disclosure relates to a distributed antenna system, and more particularly, to a method of a distributed antenna system to communicate with a base station.

#### **2. Description of the Related Art**

[0003] A distributed antenna system (DAS) may primarily serve a role of relaying macro wireless base station signals of a mobile communication service provider. The DAS is a transmission medium such as optical fiber, wired Ethernet, and the like, or a system composed of spatially separated antenna nodes (e.g., a remote unit) connected to a common node (e.g., a head-end unit) through a transmission network.

[0004] The DAS may be used to cover wide areas such as buildings, campuses, and stadiums. In particular, the DAS may be installed in an area where radio waves are not received or where radio waves are weak, such as inside buildings, underground buildings, subways, tunnels, apartment complexes in residential areas, stadiums, and the like to extend coverage of a base station by providing mobile communication services to even a shadow area where signals of the base station are difficult to reach.

[0005] An open-radio access network (O-RAN) is a radio access network based on new radio (NR) technology, a fifth-generation mobile communication system established by a 3rd generation partnership project (3GPP), and provides a standard for interoperability between distributed units and radio units manufactured by different manufacturers.

[0006] The DAS may perform communication by connecting to a base station composed of an open DU (ODU) and an open RU (ORU) provided by the O-RAN. However, because the DAS is not included in the scope of O-RAN standards, communication operations with the DAS are not considered in O-RAN standards, and thus a method for performing communication efficiently is required.

### **SUMMARY**

[0007] Provided are methods of communication between a distributed antenna system and O-RAN base stations connected to the DAS.

[0008] Provided are methods of adjusting a cell size through information exchange about DAS through communication with O-RAN base stations.

[0009] The disclosure is not limited to the above objective(s), but other objective(s) not described herein may be clearly understood by one of ordinary skill in the art from descriptions below.

[0010] According to an aspect of an inventive concept of the disclosure, a method performed by a first distributed antenna system (DAS) node unit in a communication system including a DAS, the method includes: measuring information related to a delay of at least one second DAS node unit connected to the first DAS node unit; generating delay information for the at least one second DAS

node unit based on the information related to the delay; and transmitting the delay information to an open-radio access network (O-RAN) node unit of a base station to which the DAS is connected.

[0011] In an embodiment, the method may further include receiving identification information indicating the O-RAN node unit from the O-RAN node unit.

[0012] In an embodiment, the identification information may include a serial number and an IP address of the O-RAN node unit.

[0013] In an embodiment, the method may further include receiving registration information indicating that the O-RAN node unit is connected to the first DAS node unit from the O-RAN node unit.

[0014] In an embodiment, the registration information may include a number of the base station, identification information of the O-RAN node unit, and identification information of the first DAS node unit.

[0015] In an embodiment, the identification information of the O-RAN node unit may include an IP address or serial number of the O-RAN node unit, and the identification information of the first DAS node unit may include an IP address or serial number of the first DAS node unit.

[0016] According to another aspect of an inventive concept of the disclosure, a method performed by an open-radio access network (O-RAN) node unit in a communication system including a distributed antenna system (DAS), the method includes: transmitting identification information of the O-RAN node unit to a first DAS node unit included in the DAS; receiving delay information from the first DAS node unit; and determining an antenna delay value based on the delay information.

[0017] In an embodiment, the method may further include transmitting registration information indicating that the O-RAN node unit is connected to the first DAS node unit to the first DAS node unit.

[0018] In an embodiment, the registration information may include a number of a base station including the O-RAN node unit, the identification information of the O-RAN node unit, and identification information of the first DAS node unit, the identification information of the O-RAN node unit may include an IP address or serial number of the O-RAN node unit, and the identification information of the first DAS node unit may include an IP address or serial number of the first DAS node unit.

[0019] In an embodiment, the identification information may include the serial number and IP address of the O-RAN node unit, and the O-RAN node unit may be a distributed unit of a base station that is communicatively connected to the first DAS node unit.

[0020] According to another aspect of an inventive concept of the disclosure, a method performed by a first distributed antenna system (DAS) node unit in a communication system including a DAS, the method includes: measuring information related to a delay of at least one second DAS node unit connected to the first DAS node unit; generating delay information for the at least one second DAS node unit based on the information related to the delay; and transmitting the delay information to an open-radio access network (O-RAN) node unit of a base station to which the DAS is connected through a DAS management system entity.

[0021] In an embodiment, the method may further include receiving identification information indicating the O-RAN node unit from the O-RAN node unit of the base station through the DAS management system entity.

[0022] In an embodiment, the identification information may include a serial number and an IP address of the O-RAN node unit.

[0023] In an embodiment, the method may further include receiving registration information indicating that the O-RAN node unit is connected to the first DAS node unit from the O-RAN node unit through the DAS management system entity.

[0024] In an embodiment, the registration information may include a number of the base station, identification information of the O-RAN node unit, and identification information of the first DAS

node unit.

[0025] In an embodiment, the identification information of the O-RAN node unit may include an IP address or serial number of the O-RAN node unit, and the identification information of the first DAS node unit may include an IP address or serial number of the first DAS node unit.

[0026] According to another aspect of an inventive concept of the disclosure, a method performed by an open-radio access network (O-RAN) node unit in a communication system including a distributed antenna system (DAS), the method includes: transmitting identification information of the O-RAN node unit to a DAS management system entity; receiving delay information from the DAS management system entity; and determining an antenna delay value based on the delay information.

[0027] In an embodiment, the method may further include transmitting registration information indicating that the O-RAN node unit is connected to a first DAS node unit of the DAS to the DAS management system entity.

[0028] In an embodiment, the registration information may include a number of a base station including the O-RAN node unit, the identification information of the O-RAN node unit, and identification information of the first DAS node unit, the identification information of the O-RAN node unit may include an IP address or serial number of the O-RAN node unit, and the identification information of the first DAS node unit may include an IP address or serial number of the first DAS node unit.

[0029] In an embodiment, the identification information may include the serial number and IP address of the O-RAN node unit, and the O-RAN node unit may be a distributed unit of a base station that is communicatively connected to the DAS management system entity.

[0030] According to embodiments, a cell range of a base station may be efficiently adjusted by exchanging distance information of a DAS through communication with a distributed antenna system and an O-RAN base station.

[0031] Effects obtainable by the methods according to the inventive concept are not limited to the effect(s) described above, but other effects not described herein may be clearly understood by those of ordinary skilled in the art from the above descriptions.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

[0032] These and/or other aspects will become apparent and more readily appreciated from the following description of the embodiments, taken in conjunction with the accompanying drawings in which:

[0033] FIGS. 1A and 1B are block diagrams of a communication system according to an embodiment;

[0034] FIGS. 2A to 2C are block diagrams illustrating examples of topologies of distributed antenna systems according to various embodiments;

[0035] FIGS. 3A to 3D are block diagrams of elements of a communication system according to an embodiment;

[0036] FIG. 4 is a block diagram of a communication system according to an embodiment; and

[0037] FIGS. 5 and 6 are flowcharts for explaining an operating method of the communication system illustrated in FIG. 4.

### DETAILED DESCRIPTION

[0038] In the existing communication system, a system according to open-radio access network (O-RAN) standards and a distributed antenna system (DAS) are designed to be controlled as separate systems. That is, even if the DAS is connected to the system according to the O-RAN standards, an O-RAN system is configured so that communication may be performed to a target terminal without

special consideration of this.

[0039] Therefore, because the O-RAN system is designed to control communication by considering various cases such as the connection of the DAS, an antenna delay with a sufficient value considering a maximum delay for a configuration such as a cell size is set. In this case, because the antenna delay is set to a non-optimal value and needs to be set for all sites rather than a specific site, unnecessary communication resources are wasted, communication performance is reduced, and problems such as exceeding a buffering allowance may occur depending on product characteristics.

[0040] In order to solve these problems, various aspects of the disclosure propose techniques in which an O-RAN base station determines an antenna delay by considering a delay time of a DAS section using certain information provided from a DAS.

[0041] In various embodiments, the technologies described in the disclosure and systems and devices for implementation thereof may utilize other radio access technologies such as WiFi or WiMax as well as radio access technologies such as code division multiple access (CDMA), time division multiple access (TDMA), frequency division multiple access (FDMA), orthogonal FDMA (OFDMA), single-carrier FDMA (SC-FDMA), LTE, a global system for mobile communications (GSM), 5G NR, and the like to support communication between networks (or systems).

[0042] Various other embodiments and features according to the inventive concept of the disclosure will be further described later below. It should be apparent that the teachings herein may be implemented in a wide variety of forms and any particular structure, function, or both, disclosed herein are merely exemplary, and not limiting. Based on the teachings herein, those of ordinary skill in the art will appreciate that aspects disclosed herein may be implemented independently of any other aspects, and two or more of these aspects may be combined in various ways. For example, a device or a method may be implemented by using any number of aspects set forth herein. Furthermore, the device or the method may be implemented with structures and functions of one or more of the aspects described herein, or may be implemented by using structures and functions of other aspects. For example, the method may be implemented as a part of commands stored on a non-transitory computer-readable recording medium for execution on a system, a device, an apparatus and/or a processor, or a computer. Furthermore, one aspect may include at least one component of the claim.

[0043] Hereinafter, embodiments of the disclosure will be described in detail with reference to the accompanying drawings. At this time, it should be noted that the same components in the attached drawings are indicated by the same symbols as much as possible. In addition, detailed descriptions of well-known functions and configurations that may obscure the gist of the disclosure will be omitted.

[0044] In describing the embodiments in this specification, descriptions of technical content that is well-known in the art and not directly related to the disclosure will be omitted. This is to convey the gist of the disclosure more clearly without obscuring it by omitting unnecessary explanation.

[0045] For the same reason, some components are exaggerated, omitted, or schematically shown in the accompanying drawings. In addition, the size of each component does not entirely reflect its actual size. In each drawing, identical or corresponding components are given the same reference numerals.

[0046] Advantages and features of the disclosure, and implementation methods thereof will be clarified through following embodiments described with reference to the accompanying drawings. In this regard, an embodiment of the inventive concept may have different forms and should not be construed as being limited to the descriptions set forth herein. Accordingly, the embodiments are merely described below, by referring to the figures, to explain aspects. Like reference numerals refer to like elements throughout.

[0047] At this time, it will be understood that each block of processing flow charts and combinations of the processing flow charts may be performed by computer program instructions.

Because these computer program instructions may be mounted on a processor of a general-purpose computer, special-purpose computer, or other programmable data processing equipment, the instructions performed through the processor of the computer or other programmable data processing device creates a unit to perform functions described in flow chart block(s). These computer program instructions may also be stored in computer-usable or computer-readable memory that can be directed to a computer or other programmable data processing equipment to implement the functions in a particular manner. Accordingly, the instructions stored in the computer-usable or computer-readable memory may also produce manufactured items containing an instruction unit that performs the functions described in the flow chart block(s). Because the computer program instructions can be mounted on a computer or other programmable data processing equipment, instructions that execute a computer or other programmable data processing equipment by performing a series of operations on a computer or other programmable data processing equipment to generate a computer-executable process may also provide operations for executing the functions described in the flow chart block(s).

[0048] In addition, each block may represent a module, segment, or portion of code containing one or more executable instructions for executing specified logical function(s). In addition, in some alternative implementations, it should be noted that functions mentioned in the blocks to occur out of order. For example, two blocks shown in succession may be performed substantially simultaneously, or the blocks may sometimes be performed in reverse order depending on their corresponding functions.

[0049] At this time, a term “unit” used in the present embodiment means software or hardware components such as field-programmable logic array (FPLA) and application-specific integrated circuit (ASIC), and the “unit” performs a particular function. However, the “unit” is not limited to software or hardware. The “unit” may be configured to be stored in an addressable storing medium or to play back one or more processors. Accordingly, the “unit” may include, for example, software components, object-oriented software components, components such as class components and task components, processors, formulas, attributes, procedures, subroutines, segments of program codes, drivers, firmware, micro codes, circuits, data, database, data structures, tables, arrays and variables. Functions provided in components and “units” may be combined into a smaller number of components and “units”, or may be further divided into additional components and “units.” Furthermore, components and “units” may be implemented to reproduce one or more central processing units (CPUs) within a device or a secure multimedia card.

[0050] Hereinafter, a base station is an entity that performs resource allocation for a terminal and may be at least one of a Node B, base station (BS), eNode B (eNB), gNode B (gNB), a radio access unit, a base station controller, or a node on a network. The terminal may include user equipment (UE), a mobile station (MS), a cellular phone, a smartphone, a computer, or a multimedia system capable of performing communication functions. In addition, the embodiment described below may be applied to other communication systems having a similar technical background or channel type as the embodiment. In addition, the embodiment may be applied to other communication systems through some modifications without significantly departing from the scope of the disclosure, at the discretion of one of ordinary skill in the art.

[0051] Terms used in the following description, such as terms for identifying a connection node, terms referring to network entities or network functions (NFs), terms referring to messages, terms referring to interfaces between network objects, and terms referring to various identification information, are provided as examples for convenience of explanation. Therefore, the disclosure is not limited to the terms described below, and other terms referring to objects having equivalent technical meaning may be used.

[0052] For convenience of explanation below, some terms and names defined in 3rd generation partnership project (3GPP) long-term evolution (LTE) standards may be used. However, the disclosure is not limited by the above terms and names, and may be equally applied to systems

according to other standards.

[0053] Hereinafter, various embodiments according to the inventive concept of the disclosure will be described in detail one by one.

[0054] FIGS. **1A** and **1B** are block diagrams of a communication system according to an embodiment.

[0055] Communication systems **10a** and **10b** may include O-RAN base stations **110a** and **110b**, distributed antenna systems (DASs) **120a** and **120b**, and a terminal **130**, respectively.

[0056] The communication systems **10a** and **10b** may allow radio resources, such as operating frequencies, power limits, and geographical areas, to be dynamically allocated to multiple terminals associated with the distributed antenna systems **120a** and **120b** under the control of the O-RAN base stations **110a** and **110b**.

[0057] Although the O-RAN base stations **110a** and **110b** are described as O-RAN base stations in the expression, it is obvious that base stations in a field where general communication technologies (LTE, NR, Wi-Fi, etc.) other than O-RAN are applied may also be included.

[0058] The O-RAN base stations **110a** and **110b** may include at least one of O-RAN distributed units (ODUs) **140a** and **140b**, O-RAN radio units (ORUs) **150a** and **150b**, and a management system entity (MSE) **180**, respectively. The ODU **140a** and **140b**, the ORUs **150a** and **150b**, and the MSE **180** may be electrically connected to each other. According to another embodiment, the MSE **180** may not be included in an O-RAN base station but may be configured as a separate entity. In the disclosure, it is illustrated that only ODU, ORU or the MSE **180** is included in an O-RAN base station, but it is obvious that various entities (a centralized unit (CU), RAN intelligent controller RIC, etc.) included in an actual O-RAN standard may be included. In addition, although it is expressed as an O-RAN base station in the disclosure, it may include the entire O-RAN structure and may include a base station used in general LTE, 5G, etc., not an O-RAN standard.

[0059] The O-RAN base stations **110a** and **110b** may be devices that provide radio services using any radio access technology, such as a base station, an access point, or any type of radio frequency (RF) access system.

[0060] The O-RAN base stations **110a** and **110b** may be connected to at least one distributed antenna system. In the case of FIGS. **1A** and **1B**, each of the O-RAN base stations **110a** and **110b** is illustrated as being connected to one distributed antenna system, but it is also possible for the O-RAN base stations **110a** and **110b** to be simultaneously connected to a plurality of DAS structures disclosed in FIGS. **2A** to **2C**.

[0061] Each of distances between the ODU **140a** and **140b** and the ORUs **150a** and **150b** in the O-RAN base stations **110a** and **110b** may be defined as a fronthaul distance **190**. In general, the fronthaul distance **190** may be determined as a distance between an ODU and an ORU.

[0062] The O-RAN base stations **110a** and **110b** may determine a cell size **195** for performing communication. In general, a cell size may be set to a certain distance from the O-RAN base stations **110a** and **110b**. The cell size **195** may be subject to constraints based on random access channel (RACH) signal processing standards or guard time constraints in a case of time division duplex (TDD). For example, a cell size may be determined to be 15 km for LTE and 2 to 5 km for 5G NR according to the RACH signal processing standards. In addition, in the case of TDD, the guard time constraints may be determined to be 12 km for LTE and 5 km for 5G NR. Depending on the determined cell size **195**, a configuration inside the DAS may be restricted. That is, depending on constraints on a cell size, when a distributed antenna system is connected to an O-RAN base station, an allowable distance between a DAS headend unit and a DAS remote unit may be limited.

[0063] The O-RAN base stations **110a** and **110b** may determine an antenna delay value for performing communication with the terminal **130**. For example, when the ODU **140a** and **140b** determine a fixed value corresponding to a delay of a distributed antenna system section and transmit the fixed value to the ORUs **150a** and **150b** as a parameter value (e.g., a tda parameter value), the ORUs **150a** and **150b** may reduce a buffering delay by the corresponding parameter

value. However, because the conventional O-RAN base station does not transmit or receive special information with a distributed antenna system in relation to a corresponding delay value, an antenna delay value may be determined arbitrarily without considering a situation of the distributed antenna system connected to the conventional O-RAN base station. That is, depending on an installation environment of a distributed antenna system, delays of sections of a plurality of distributed antenna systems may be different, but because information about each of a plurality of distributed antenna systems connected to the ORUs **150a** and **150b** is not received, the same antenna delay value is set for all distributed antenna systems.

[0064] When an O-RAN base station does not interoperate with a distributed antenna system, if the same antenna delay value is set for all ORUs, a maximum delay value for each site of a distributed antenna system section measured by a headend unit of the distributed antenna system is not reflected, so the O-RAN base station needs to determine an antenna delay value that is sufficiently large to cover all delay values. In this case, the type and characteristics of a product installed may differ depending on the site (e.g., a buffering allowance for delay compensation), and a difference in a maximum distance between a headend unit and a DAS remote unit may be large. That is, by applying a sufficiently large delay value to all sites, a problem that exceeds an allowable range of buffering may occur depending on product characteristics, which may result in poor communication quality and waste of resources. In order to solve the above problem, it will be described in detail below.

[0065] Referring to FIGS. **1A** and **1B**, the ORUs **150a** and **150b** may be communicatively connected to the ODUs **140a** and **140b**, an MSE **180a**, and headend units **160a** and **160b**. In addition, the ORUs **150a** and **150b** may be electrically connected to the ODUs **140a** and **140b** and the MSE **180a**, and may be physically connected to headend units **160a** and **160b**. For example, the ORUs **150a** and **150b** may be connected to the headend units **160a** and **160b** via a copper wire.

[0066] The ORUs **150a** and **150b** may be named a DAS interface unit DIU.

[0067] According to an embodiment, the ORUs **150a** and **150b** may transmit an ORU identifier and information indicating which headend unit an ORU is connected to to the headend units **160a** and **160b** or the MSE **180a**. The ORU identifier may include information such as a serial number, IP address, and ORU ID of the ORUs **150a** and **150b**. Information indicating which headend unit an ORU is connected to may be represented as ORU registration information. For example, the ORU registration information may include an SITE number (or a base station number), an IP address of the ORU, and an IP address of a headend. That is, the registration information may be represented as {Site #17, ORU #110.20.1.8, HE #110.20.2.1}.

[0068] According to an embodiment, the ORUs **150a** and **150b** may receive delay information from the headend units **160a** and **160b** or an MSE **180b** to farthest DAS remote units **170a** and **170b** measured by the headend units **160a** and **160b**. The ORUs **150a** and **150b** may determine an antenna delay value (e.g., an external antenna delay) based on the received delay information.

[0069] In an embodiment, the O-RAN base stations **110a** and **110b** may determine various parameters for performing communication based on distance information (or delay information) received from the DASs **120a** and **120b**. For example, by measuring a starting point of a cell differently, an effect of expanding a size of the entire cell may be achieved even if the cell size is limited. This will be described in more detail with reference to FIG. **4** below.

[0070] Referring to FIG. **1B**, the communication system **10b** may include the MSE **180a**. The MSE **180a** may be communicatively connected to the ORU **150b** and the headend unit **160b**. The MSE **180a** may be included in the O-RAN base station **110b** or the distributed antenna system **120b**. The MSE **180a** may be configured as a separate entity physically distinct from the O-RAN base station **110b** or the distributed antenna system **120b**.

[0071] According to an embodiment, the MSE **180a** may receive an ORU identifier and information indicating which headend unit an ORU is connected to from the ORU **150b**. The MSE **180a** may forward the received ORU identifier and information indicating which headend unit an

ORU is connected to a headend unit. The ORU identifier may include information such as a serial number, IP address, and ORU ID of the ORUs **150a** and **150b**. Information indicating which headend unit an ORU is connected to may be represented as ORU registration information. For example, the ORU registration information may include a SITE number (or base station number), an IP address of the ORU, and an IP address of a headend. That is, the registration information may be represented as {Site #17, ORU #110.20.1.8, HE #110.20.2.1}.

[0072] According to an embodiment, the MSE **180a** may receive delay information from the headend unit **160b** to the farthest DAS remote unit **170b** measured by the headend unit **160b** and transmit the delay information to the ORU **150b**.

[0073] The MSE **180a** may be an entity for integrated management and operation of the distributed antenna systems **120a** and **120b** as described above, but is not limited thereto. In an embodiment, the MSE **180a** may be a network management system or a DAS management system provided by a manufacturer of the distributed antenna system **120b**. In another embodiment, the MSE **180a** may be a network management system provided by a manufacturer of the O-RAN base station **110b**.

[0074] The distributed antenna systems **120a** and **120b** may combine or distribute wireless communication services provided by the O-RAN base stations **110a** and **110b** and provide them to end user terminals within coverage.

[0075] According to an embodiment, the distributed antenna systems **120a** and **120b** may include the headend units (HEUs) **160a** and **160b** connected to the O-RAN base stations **110a** and **110b**, and remote units (RUs) (or DAS RUs) **170a** and **170b** connected to the headend units (HEUs) **160a** and **160b** in a point-to-multipoint structure.

[0076] The distributed antenna systems **120a** and **120b** may provide radio services from the O-RAN base stations **110a** and **110b** to end user terminals by using radio resources allocated under direct or indirect control of a system controller. The distributed antenna systems **120a** and **120b** may be configured in various structures, which will be described in detail in FIGS. 2A to 2C.

[0077] Referring to FIGS. 1A and 1B, the headend units **160a** and **160b** may be connected to the O-RAN base stations **110a** and **110b**. The headend units **160a** and **160b** may be connected to the O-RAN base stations **110a** and **110b** by wire.

[0078] Referring to FIG. 1A, the headend unit **160a** may be communicatively connected to the ORU **150a** of the O-RAN base station **110a** and at least one DAS remote unit **170a**. The ORU **150a** may be connected to the headend unit **160a** by wire. For example, they may be connected to each other via a copper wire. Referring to FIG. 1B, the headend unit **160a** may be communicatively connected to the MSE **180a**.

[0079] According to an embodiment, the headend units **160a** and **160b** may receive an ORU identifier and information indicating which headend unit an ORU is connected to from the ORUs **150a** and **150b**. The ORU identifier may include information such as a serial number, IP address, and ORU ID of the ORUs **150a** and **150b**. Information indicating which headend unit an ORU is connected to may be represented as ORU registration information. For example, the ORU registration information may include a SITE number (or base station number), an IP address of the ORU, and an IP address of a headend. That is, the registration information may be represented as {Site #17, ORU #110.20.1.8, HE #110.20.2.1}.

[0080] According to an embodiment, the headend units **160a** and **160b** may receive an ORU identifier and information indicating which headend unit an ORU is connected to from the ORUs **150a** and **150b** through the MSE **180a**.

[0081] According to an embodiment, the headend units **160a** and **160b** may measure a delay (or delay time) to the DAS remote units **170a** and **170b** located farthest from at least one connected DAS remote units **170a** and **170b** connected to the headend units **160a** and **160b**. The headend units **160a** and **160b** may generate delay information (or delay time information) based on the measured delay (or delay time) and transmit it to the ORUs **150a** and **150b**. Alternatively, the headend units **160a** and **160b** may transmit the delay information (or delay time information) to the

ORUs **150a** and **150b** through the MSE **180a** based on the measured delay (or delay time).

[0082] The DAS remote units **170a** and **170b** of FIG. **1A** and FIG. **1b**, unlike a remote radio head, which is an RF processing device of a distributed base station, may integrate and process multiple radio services, and accordingly, in FIG. **1**, only an embodiment in which the DAS remote units **170a** and **170b** are connected to one of the O-RAN base stations **110a** and **110b** is illustrated, but the DAS remote units **170a** and **170b** may be connected directly to a plurality of radio service devices (e.g., base stations) or with a certain network interposed therebetween.

[0083] The DAS remote units **170a** and **170b** may include at least one antenna (not shown) connected thereto by wire. For example, the DAS remote units **170a** and **170b** may be connected to the antenna by a copper wire. The DAS remote units **170a** and **170b** may be communicatively connected to the headend units **160a** and **160b**.

[0084] According to an embodiment, the DAS remote units **170a** and **170b** may be configured to cause the headend units **160a** and **160b** to perform an operation for measuring a delay (or delay time).

[0085] The terminal **130** of FIGS. **1A** and **1B** illustrates user equipment (UE) located at an edge portion according to a determined cell size of the O-RAN base stations **110a** and **110b**. Therefore, in general, a distance from the ORUs **150a** and **150b** to the UE **130** may be determined as a cell size. The cell structure may be determined in various ways depending on the setting, but is generally determined as a circle, and the cell size may represent a radius of the cell.

[0086] The UE **130** may receive communication data transmitted through the antenna connected to the DAS remote units **170a** and **170b**.

[0087] According to the inventive concept of the disclosure, elements of the communication systems **10a** and **10b**, that is, the number of node units (headend units, remote units, MSEs, etc.) constituting a distributed antenna system, and their connection topologies are not limited to the embodiment shown in FIG. **1**, and various modifications and variations are possible.

[0088] FIGS. **2A** to **2C** are block diagrams illustrating examples of topologies of distributed antenna systems according to various embodiments. Headend units **260a** to **260c** and DAS remote units **270a** to **270k** described in FIGS. **2A** to **2C** may be the same as the headend units **160a** and **160b** and the DAS remote units **170a** and **170b** described in FIG. **1**.

[0089] Referring to FIGS. **2A** to **2C**, distributed antenna systems **20a** to **20c** may include the headend units **260a** to **260c** and the DAS remote units **270a** to **270k**. According to another embodiment, the distributed antenna systems **20a** to **20c** may also include a DAS management system (not shown).

[0090] Referring to FIG. **2A**, the first distributed antenna system **20a** may include the headend unit **260a**, the remote unit **270a** connected to the headend unit **260a**, and the DAS remote unit **270b** connected to the remote unit **270a** in a daisy chain structure or a cascade structure. The DAS remote units **270a** and **270b** may be capable of integrated processing of multiple radio services, and accordingly, although FIG. **2A** only illustrates an embodiment in which the DAS remote units **270a** and **270b** are connected to one headend unit **260a**, they may be connected directly to a plurality of headend units or with a certain network unit interposed therebetween.

[0091] Referring to FIG. **2B**, the second distributed antenna system **20b** may include the headend unit **260b** and the DAS remote units **270c** to **270e** connected to the headend unit **260b** in a point-to-multipoint structure.

[0092] Referring to FIG. **2C**, the third distributed antenna system **20c** may include the headend unit **260c**, DAS remote units **270f** to **270i** connected to a headend unit **130a** in a point-to-multipoint structure, and the DAS remote units **270h**, **270j**, and **270k**, each of which is connected to a corresponding remote unit among the DAS remote units **270g** and **270i** in a daisy chain structure or a cascade structure.

[0093] Although not shown, the distributed antenna systems **20a** to **20c** may optionally further include a DAS management system, and the DAS management system may be communicatively

connected to the headend units **260a** to **260c**.

[0094] The distributed antenna systems **20a** to **20c** may provide radio services to end user devices by using radio resources allocated under direct or indirect control from an O-RAN base station.

[0095] According to the inventive concept of the disclosure, the number of node units (headend units and DAS remote units) constituting the distributed antenna systems **20a** to **20c** and their connection topologies are not limited to the examples shown in FIGS. **2A** to **2C**, and various changes and modifications are possible.

[0096] FIGS. **3A** to **3D** are block diagrams of elements of a communication system according to an embodiment. In describing FIGS. **3A** to **3D**, because the same reference numerals as in FIGS. **1A** and **1B** represent the same reference numerals, FIGS. **1A** and **1B** are referred to together, but redundant descriptions are omitted.

[0097] Referring to FIG. **3A**, the ORU **150** included in an O-RAN base station may include an ORU processor **305** and an ORU interface **310**.

[0098] The ORU processor **305** may include, although not shown, a controller that controls all procedures to perform an operation of an ORU in a communication system according to an embodiment, and a converter for converting service signals.

[0099] The ORU processor **305** may generate an ORU identifier and information indicating which headend unit an ORU is connected to.

[0100] The ORU processor **305** may determine an antenna delay value based on delay information (or delay time information) received from the MSE **180** or a headend unit **160**.

[0101] The ORU interface **310** is for the ORU **150** to transmit and receive information necessary for improving cell size constraints to and from an ODU **140**, the headend unit **160**, and the MSE **180**.

[0102] The ORU interface **310** may transmit an ORU identifier and information indicating which headend unit an ORU is connected to to the headend unit **160** or the MSE **180**.

[0103] That is, the ORU **150** may transmit and receive the information to and from the ODU **140** connected through a first communication link **CL1a**, and the headend unit **160** and the MSE **180** connected through second communication links **CL2a** and **CL2b**, respectively, by using the ORU interface **310**.

[0104] Here, the first communication link **CL1a** and the second communication links **CL2a** and **CL2b** may be, for example, the Internet, but are not limited thereto. Alternatively, the first communication link **CL1a** and the second communication links **CL2a** and **CL2b** may be media for transmitting analog or digital type service signals. The first communication link **CL1a** and the second communication links **CL2a** and **CL2b** may be any wired and/or wireless communication link such as WiMax, a network optical fiber, an Ethernet-based cable, a wired cable formed of copper, an optical link, etc.

[0105] Referring to FIG. **3B**, the headend unit **160** may include a headend unit processor **315** (hereinafter referred to as an HEU processor) and a headend unit interface **320** (hereinafter referred to as an HEU interface).

[0106] The HEU interface **320** is for the headend unit **160** to transmit and receive information necessary for improving cell size constraints to and from the ORU **150**, a DAS remote unit **170**, and the MSE **180**.

[0107] The HEU interface **320** may receive an ORU identifier and information indicating which headend unit an ORU is connected to from the ORU **150** or the MSE **180**.

[0108] The HEU interface **320** may transmit delay information generated based on a measured delay (or delay time) to the farthest DAS remote unit **170** to the ORU **150**.

[0109] The headend unit **160** may transmit the information described above to the O-RAN base stations **110a** and **110b** using a certain security protocol, for example, an HTTPS protocol.

[0110] The headend unit **130** may transmit and receive pieces of information such as allocation information to and from the DAS remote unit **170** and the MSE **180** by using the security protocol

described above or another security protocol defined by a manufacturer of a distributed antenna system.

[0111] The headend unit **160** may transmit and receive the pieces of information to and from the MSE **180** and the ORU **150** connected through a first communication link CL**1b** and the second communication link CL**2a**, respectively, and the DAS remote unit **170** connected through a fifth communication link CL**5**, by using the HEU interface **320**.

[0112] The fifth communication link CL**5** may be, for example, the Internet, but is not limited thereto, and may be a medium for transmitting an analog or digital type service signal. The fifth communication link CL**5** may be any wired and/or wireless communication link such as an RF cable, WiMax, etc., a network optical fiber, an Ethernet-based cable, a wired cable formed of copper, an optical link, etc.

[0113] The HEU processor **315** may include, although not shown, a controller that controls all procedures to perform an operation of a headend unit in a communication system according to an embodiment, and a converter for converting service signals.

[0114] The HEU processor **315** may measure a delay (or delay time) to the DAS remote unit **170** located farthest from at least one connected DAS remote unit **170**. The HEU processor **315** may generate delay information (or delay time information) based on the measured delay (or delay time).

[0115] Referring to FIG. **3C**, the DAS remote unit **170** may include a DAS remote unit interface **330** (hereinafter referred to as a DAS RU interface) and a remote unit processor **325** (hereinafter referred to as a DAS RU processor).

[0116] The DAS RU interface **330** is for transmitting and receiving information required for communicating to and from the headend unit **160**, the MSE **180**, and another DAS remote unit.

[0117] The DAS remote unit **170** may transmit and receive the above information to and from the headend unit **160** and the MSE **180** by using an HTTPS protocol and other security protocols, according to an embodiment.

[0118] The DAS remote unit **170** may transmit and receive the above information to and from the MSE **180**, the headend unit **160**, and another remote unit through a first communication link CL**1c**, the fifth communication link CL**5**, and a sixth communication link CL**6**, respectively, by using the DAS RU interface **330**.

[0119] The sixth communication link CL**6** may be, for example, the Internet, but is not limited thereto, and may be a medium for transmitting an analog or digital type service signal. The sixth communication link CL**6** may be any wired and/or wireless communication link such as an RF cable, WiMax, etc., a network optical fiber, an Ethernet-based cable, a wired cable formed of copper, an optical link, etc.

[0120] The DAS RU processor **325** may include, although not shown, a controller that controls all procedures to perform an operation of the DAS remote unit **170** in a communication system according to an embodiment, and a converter for converting service signals.

[0121] The DAS RU processor **325** may perform analog and/or digital filtering, amplification, and other processing on received service signals and then transmit them to an end user terminal via a connected antenna.

[0122] Referring to FIG. **3D**, the MSE **180** may include an MSE interface **335**, a bus **340**, and an MSE processing system **345**.

[0123] The MSE interface **335** is for the MSE **180** to transmit and receive information necessary for improving cell size constraints to and from the headend unit **160**, the DAS remote unit **170**, and the ORU **150**.

[0124] The MSE **180** may transmit and receive the information to and from the headend unit **160**, the DAS remote unit **170**, and the ORU **150** connected to the MSE **180** through the first communication links CL**1b** and CL**1c** and the second communication link CL**2b**, respectively, by using the MSE interface **335**.

[0125] The bus **340** may communicatively connect the MSE interface **335** to the MSE processing system **345**.

[0126] The MSE processing system **345** may include a processor **350** and a memory **355**.

[0127] The processor **350** may include, although not shown, a controller that controls all procedures to perform an operation of the MSE **180** in a communication system according to an embodiment, and a converter for converting service signals.

[0128] The processor **350** may be any device suitable for executing a program command for processing various information or delay time measurements received from the O-RAN base stations **110a** and **110b** or a DAS through the MSE interface **335**.

[0129] The processor **350** may be any device suitable for executing program commands for monitoring, managing, controlling, and operating all operating states of the O-RAN base stations **110a** and **110b**, or program commands for monitoring, managing, controlling, and operating all operating states of the DASs **120a** and **120b**.

[0130] The memory **355** may be any non-transitory medium for storing the program commands described above defining an operation of the MSE **180**. For example, the memory **355** may be read-only memory (ROM), random-access memory (RAM), an optical storage, a magnetic storage, a flash memory, or any other medium.

[0131] FIG. **4** is a block diagram of a communication system according to an embodiment, and FIGS. **5** and **6** are flowcharts for explaining an operating method of the communication system illustrated in FIG. **4**.

[0132] In more detail, a communication system **10c** illustrated in FIG. **4** exemplifies an embodiment in which a headend unit **160c** of a distributed antenna system **120c** is interoperable with an ORU **150c** of an O-RAN base station **110c**, and the ORU **150c** and the headend unit **160c** are communicatively connected to each other so that they may transmit and receive information necessary to determine an antenna delay value through their respective interfaces. Alternatively, an embodiment may be exemplified in which the ORU **150c** and the headend unit **160c** are communicatively connected to each other so that they may transmit and receive necessary information through the MSE **180b**.

[0133] In FIG. **4**, between the O-RAN base station **110c** and the headend unit **160c**, and among the headend unit **160c**, a DAS remote unit **170c**, and the MSE **180b** constituting the distributed antenna system **120c**, radio service signals provided to/from an end user terminal are transmitted as analog or digital type signals. Processes related to configurations for this will not be given herein for convenience of explanation.

[0134] In the description of FIGS. **4** to **6**, the same or corresponding reference numerals as those in FIGS. **1** to **3D** denote the same or corresponding elements, and therefore, repeated descriptions thereof will not be given herein. In a communication system according to the present embodiment, signal transmission and reception operations for an O-RAN base station and a distributed antenna system including the ORU **150**, the headend unit **160**, the DAS remote unit **170c**, and the MSE **180** will be mainly described.

[0135] Referring to FIG. **4**, a series of operations in the O-RAN base station **110c** and the distributed antenna system **120c** according to an embodiment may be performed to set a distance from an ODU **140c** to a DAS remote unit **170c** as a fronthaul distance **410** based on an antenna delay value determined by the ORU **150c**, and a distance from the DAS remote unit **170c** to the UE **130** as a cell size **420**. In this case, even under the same cell constraints as before, an effect of expanding an actual cell range may occur. The above series of operations will be described in detail in FIGS. **5** and **6** below.

[0136] Referring to FIG. **5**, first, the ORU **150** of an O-RAN base station may generate identification information of an ORU and registration information of an ORU. The ORU identifier may include information such as a serial number, an IP address, and RU ID of the ORU **150**, a geographical location of an entity, and an installation location of an antenna. Registration

information of the ORU may include information indicating which headend unit the ORU is connected to. For example, the registration information may include an SITE number (or a base station number), identifier information of the ORU (IP address, serial number, etc.), and identifier information of the headend unit (IP address, serial number, etc.). That is, the registration information may be represented as {Site #17, ORU #110.20.1.8, HE #110.20.2.1}.

[0137] Identification information generated by the ORU **150** may be considered as identification information of an O-RAN base station including the ORU **150** or an ODU connected to the ORU.

[0138] The ORU identification information may further include an identifier indicating interoperability of the ORU **150** and the headend unit **160** of a distributed antenna system. For example, the ORU identification information may include an indicator indicating that delay information is to be transmitted because the ORU **150** interoperates with the headend unit **160**.

[0139] In operation **S501**, the ORU **150** may transmit generated ORU identification information to the headend unit **160**.

[0140] In addition, in operation **S502**, the ORU **150** may transmit generated ORU registration information to the headend unit **160**.

[0141] The headend unit **160** that receives ORU identification information or/and ORU registration information may identify an interoperable ORU **150** based on the ORU identification information or the ORU registration information. In addition, the headend unit **160** may determine whether to interoperate with the ORU **150**.

[0142] In operation **S503**, the headend unit **160** may measure a delay (or time delay) for at least one DAS remote unit **170** connected to the headend unit **160** in a distributed antenna system. A method by which the headend unit **160** measures the delay (or time delay) for the DAS remote unit **170** may be a conventionally known method such as measuring a round trip time or reporting a communication quality measurement, and this will not be described in detail.

[0143] In operation **S504**, the headend unit **160** may identify a farthest DAS remote unit **170** based on the measured delay (or delay time) for at least one connected DAS remote unit **170**, and may generate delay information (or distance information or delay time information) based on the information.

[0144] In operation **S505**, the headend unit **160** may transmit the generated delay information to the interoperable ORU **150**.

[0145] In operation **S506**, the ORU **150** that receives the delay information may determine an antenna delay value (e.g., an external antenna delay value) based on the delay information. Once the antenna delay value is determined, the ORU **150** may reduce a buffering delay by the antenna delay value. Therefore, it can be set to have a widest cell size by considering the furthest DAS remote unit **170** and reducing a buffering delay.

[0146] Referring to FIG. **6**, first, the ORU **150** of an O-RAN base station may generate identification information of an ORU and registration information of an ORU. The ORU identifier may include information such as a serial number, an IP address, and an ORU ID of the ORU **150**, a geographical location of an entity, and an installation location of an antenna. Registration information of the ORU may include information indicating which headend unit the ORU is connected to. For example, the registration information may include an SITE number, identifier information of the ORU (IP address, serial number, etc.), and identifier information of the headend unit (IP address, serial number, etc.). That is, the registration information may be represented as {Site #17, ORU #110.20.1.8, HE #110.20.2.1}.

[0147] Identification information generated by the ORU **150** may be considered as identification information of an O-RAN base station including the ORU **150** or an ODU connected to the ORU.

[0148] The ORU identification information may further include an identifier indicating interoperability of the ORU **150** and the headend unit **160** of a distributed antenna system. For example, the ORU identification information may include an indicator indicating that delay information is to be transmitted because the ORU **150** interoperates with the headend unit **160**.

[0149] In operation **S601**, the ORU **150** may transmit generated ORU identification information to the MSE **180**.

[0150] In addition, in operation **S602**, the ORU **150** may transmit generated ORU identification information to the MSE **180**.

[0151] The headend unit **160** that receives ORU identification information or/and ORU registration information from the ORU **150** may identify an interoperable ORU **150** based on the ORU identification information or ORU registration information. Further, the MSE **180** may determine whether to interoperate with the ORU **150**.

[0152] In operation **S603**, the MSE **180** may transmit the received ORU identification information or ORU registration information to the headend unit **160** based on headend unit identifier information included in the ORU registration information.

[0153] In operation **S604**, the headend unit **160** that receives the ORU identification information or/and ORU registration information may measure a delay (or time delay) for at least one DAS remote unit **170** connected to the headend unit **160** in a distributed antenna system. A method by which the headend unit **160** measures the delay (or time delay) for the DAS remote unit **170** may be a conventionally known method such as measuring a round trip time or reporting a communication quality measurement, and this will not be described in detail.

[0154] In operation **S605**, the headend unit **160** may identify a farthest DAS remote unit **170** based on the measured delay (or delay time) for at least one connected DAS remote unit **170**, and may generate delay information (or distance information, delay time information) based on the information.

[0155] In operation **S606**, the headend unit **160** may transmit the generated delay information to the interoperable ORU **150** through the MSE **180**.

[0156] In operation **S607**, the ORU **150** that receives the delay information from the MSE **180** may determine an antenna delay value (e.g., an external antenna delay value) based on the delay information. Once the antenna delay value is determined, the ORU **150** may reduce a buffering delay by the antenna delay value. Therefore, it can be set to have a widest cell size by considering the furthest DAS remote unit **170** and reducing a buffering delay.

[0157] FIGS. **4** and **5** describe the embodiment in which the headend unit **160** interoperates with the ORU **150** of an O-RAN base station above. However, even in an embodiment in which at least one DAS remote unit **170** interoperates with at least one radio service device, operations such as those in FIGS. **4** to **6** will be possible.

[0158] Further, the methods described with reference to FIGS. **4** to **6** include one or more operations or actions for achieving the methods. The operations and/or actions for achieving the methods may be interchanged with one another without departing from the scope of the claims. In other words, the order and/or use of specific operations and/or actions may be modified without departing from the scope of the claims, unless a certain order for the operations and/or actions is specified.

[0159] In addition, various operations of the methods described above may be performed by any suitable means capable of performing corresponding functions. The means includes, but is not limited to, various hardware and/or software components and/or modules such as an ASIC or a processor. In general, when there are operations corresponding to the drawings, these operations may have a corresponding counterpart and functional components having the same number as the number of the counterpart.

[0160] The various illustrative logic blocks, components, or circuits described in connection with the disclosure may be implemented or performed by a general-purpose processor designed to perform the functions disclosed herein, a digital signal processor (DSP), an ASIC, a field-programmable gate array (FPGA) or other programmable logic device (PLD), a discrete gate or transistor logic device, discrete hardware components, or any combination thereof. The general-purpose processor may be a microprocessor, but may alternatively be any commercially available

processor, controller, microcontroller, or state machine. The processor may also be implemented in a combination of computing devices, for example, a combination of the DSP and the microprocessor, a plurality of microprocessors, one or more microprocessors in connection with a DSP core, or any other configuration.

[0161] The term “determine” includes a wide variety of actions. For example, the term “determine” may include computing, processing, deriving, examining, looking up (e.g., looking up in a table, database, or other data structure), identifying, and the like. The term “determine” may also include receiving (e.g., receiving information), accessing (accessing data in a memory), and the like. The term “determine” may also include resolving, selecting, choosing, establishing, and the like.

[0162] Numerous modifications and adaptations will be readily apparent to one of ordinary skill in the art without departing from the spirit and scope of the disclosure.

[0163] Accordingly, the embodiments illustrated in the disclosure are not intended to limit the inventive concept of the disclosure but are for illustrative purposes, and the scope of the inventive concept of the disclosure is not limited by these embodiments.

[0164] The scope of protection of the inventive concept of the disclosure should be interpreted in accordance with the claims below, and all technical ideas within the equivalent scope should be construed as being included in the scope of inventive concept of the disclosure.

## Claims

1. A method performed by a first distributed antenna system (DAS) node unit in a communication system including a DAS, the method comprising: measuring information related to a delay of at least one second DAS node unit connected to the first DAS node unit; generating delay information for the at least one second DAS node unit based on the information related to the delay; and transmitting the delay information to an open-radio access network (O-RAN) node unit of a base station to which the DAS is connected.
2. The method of claim 1, further comprising: receiving identification information indicating the O-RAN node unit from the O-RAN node unit.
3. The method of claim 2, wherein the identification information comprises a serial number and an IP address of the O-RAN node unit.
4. The method of claim 1, further comprising: receiving registration information indicating that the O-RAN node unit is connected to the first DAS node unit from the O-RAN node unit.
5. The method of claim 4, wherein the registration information comprises a number of the base station, identification information of the O-RAN node unit, and identification information of the first DAS node unit.
6. The method of claim 5, wherein the identification information of the O-RAN node unit comprises an IP address or serial number of the O-RAN node unit, and the identification information of the first DAS node unit comprises an IP address or serial number of the first DAS node unit.
7. A method performed by an open-radio access network (O-RAN) node unit in a communication system including a distributed antenna system (DAS), the method comprising: transmitting identification information of the O-RAN node unit to a first DAS node unit included in the DAS; receiving delay information from the first DAS node unit; and determining an antenna delay value based on the delay information.
8. The method of claim 7, further comprising: transmitting registration information indicating that the O-RAN node unit is connected to the first DAS node unit to the first DAS node unit.
9. The method of claim 8, wherein the registration information comprises a number of a base station including the O-RAN node unit, the identification information of the O-RAN node unit, and identification information of the first DAS node unit, the identification information of the O-RAN node unit comprises an IP address or serial number of the O-RAN node unit, and the identification

information of the first DAS node unit comprises an IP address or serial number of the first DAS node unit.

**10.** The method of claim 7, wherein the identification information comprises the serial number and IP address of the O-RAN node unit, and the O-RAN node unit is a distributed unit of a base station that is communicatively connected to the first DAS node unit.

**11.** A method performed by a first distributed antenna system (DAS) node unit in a communication system including a DAS, the method comprising: measuring information related to a delay of at least one second DAS node unit connected to the first DAS node unit; generating delay information for the at least one second DAS node unit based on the information related to the delay; and transmitting the delay information to an open-radio access network (O-RAN) node unit of a base station to which the DAS is connected through a DAS management system entity.

**12.** The method of claim 11, further comprising: receiving identification information indicating the O-RAN node unit from the O-RAN node unit of the base station through the DAS management system entity.

**13.** The method of claim 12, wherein the identification information comprises a serial number and an IP address of the O-RAN node unit.

**14.** The method of claim 11, further comprising: receiving registration information indicating that the O-RAN node unit is connected to the first DAS node unit from the O-RAN node unit through the DAS management system entity.

**15.** The method of claim 14, wherein the registration information comprises a number of the base station, identification information of the O-RAN node unit, and identification information of the first DAS node unit.

**16.** The method of claim 15, wherein the identification information of the O-RAN node unit comprises an IP address or serial number of the O-RAN node unit, and the identification information of the first DAS node unit comprises an IP address or serial number of the first DAS node unit.

**17.** A method performed by an open-radio access network (O-RAN) node unit in a communication system including a distributed antenna system (DAS), the method comprising: transmitting identification information of the O-RAN node unit to a DAS management system entity; receiving delay information from the DAS management system entity; and determining an antenna delay value based on the delay information.

**18.** The method of claim 17, further comprising: transmitting registration information indicating that the O-RAN node unit is connected to a first DAS node unit of the DAS to the DAS management system entity.

**19.** The method of claim 18, wherein the registration information comprises a number of a base station including the O-RAN node unit, the identification information of the O-RAN node unit, and identification information of the first DAS node unit, the identification information of the O-RAN node unit comprises an IP address or serial number of the O-RAN node unit, and the identification information of the first DAS node unit comprises an IP address or serial number of the first DAS node unit.

**20.** The method of claim 17, wherein the identification information comprises the serial number and IP address of the O-RAN node unit, and the O-RAN node unit is a distributed unit of a base station that is communicatively connected to the DAS management system entity.

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